Date: 31/10/2026

MON TUE WED THS FRI SAT

Assignment -1

Analysis of Algorithms

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Q1:

1- Algorithmic Strategy

Initialize smallest = A[0], largest → A[0]

for i → 1 to n-1 do
if A[i] < smallest then

smallest → A[i]
end if
if A[i] > largest then

largest → A[i]
end if
end for

return smallest, largest

2- Primitive Operations:

- Comparison
- Assignment
- Increment

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Best Case: O(n)

Space Complexity:

Worst Case: O(n)

O(1)

Q2:

Algorithm 1: linear Search

Pseudocode:

```
for i -> 0 to n-1 do
  if A[i] = x then
    return TRUE
  end if
end for
return FALSE
```

Best Case: O(1)

Space complexity: O(1)

Worst Case: O(n)

→

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Algorithm 2: Binary Search
condition: Array must be sorted

pseudocode:
//

low → 0
high → n-1

while low ≤ high do
mid → (low + high) / 2

if A[mid] = X then
return TRUE
else if X < A[mid] then
high → mid-1
else
low → mid+1
end if
end while
return FALSE //
Best case: O(1)
Worst case: O(log n)
space: O(1)

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Comparisan:

Property	Linear-Search	Binary-Search
Sorted Array Required	No	yes
Best Case	O(1)	O(1)
Average/Worst case	O(n)	O(logn)
Suitable for Unsorted Data	yes	No
Suitable for Large Sorted	Less efficient	More efficient

- Linear Search works on unsorted arrays, and is simple; but Binary search is faster for large arrays because it reduces the space by half at each step...